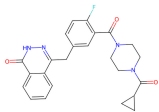
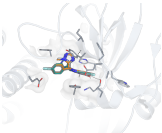


1 Drug set



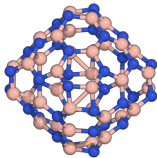
30 anti-TNBC drugs
3 families
PubChem structures

2 PARP1 docking



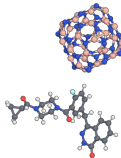
PDB 4UND, chain A
Vina 1.2.7
redock RMSD 0.5–1.3 Å

3 B₃₆N₃₆ cage



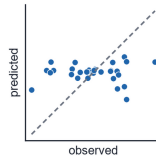
GP(1,1) BN fullerene
108 B–N bonds
GFN2-xTB minimum

4 Adsorption



4 relaxed poses per drug
 ΔE_{int} , ΔE_{ads}
bond-integrity check

5 QSPR



4 descriptors, ridge
nested 5×5 CV
Y-scrambling, AD

Outcome

18 physisorbed
12 chemisorbed
Vina $Q_{\text{CV}}^2 = -0.08$
 ΔE_{int} $Q_{\text{CV}}^2 = -0.08$